

Workflow: Building a density surface

A GRID

`vect() %>% rast(resolution = )`

an empty raster over the area, in a projected CRS

then

COUNTS

`rasterize(fun = "length", background = 0)`

the records in each cell; zero, not NA, where none fell

then

A KERNEL

`focalMat(d = , type = "Gauss")`

weights that sum to one; d is the bandwidth, chosen from the scale of the question and reported with the map

then

THE SURFACE

`focal(w = kernel, fun = "sum", na.rm = TRUE) / cell area`

records per cell, then per square kilometer; the sum is the count less the weight that reached past the edges

then

READ IT

`where.max() xyFromCell() terra::extract() mask()`

locate the peak, read the density at points, blank the cells outside the area; build the surface from observers to test whether a peak describes effort